NUMBER SYSTEMS & DATA REPRESENTATION

REPRESENTING INFORMATION IN COMPUTERS

- All different types of information in **computers** can be represented by **binary code**.

  - **Numbers**
  - **Letters of the alphabet and special characters**
  - **Microprocessor Instruction.**
  - **Graphics/Video**
  - **Sound**

NUMBER SYSTEM

- In a number, the position of digit starts from the right-hand side of the number. The rightmost digit has position 0, the next digit on its left has position 1, and so on. The digits of a number have two kinds of values:
  1. **Face value, and**
  2. **Position value.**
- The face value of a digit is the digit located at that position. For example, in decimal number 52, face value at position 0 is 2 and face value at position 1 is 5.
- The position value of a digit is ($base^{position}$). For example, in decimal number 52, the position value of digit 2 is 10^0 and the position value of digit 5 is 10^1 . Decimal numbers have a base of 10.

NUMBER SYSTEM

- The number is calculated as the:
  - sum of (face value * $base^{position}$), of each of the digits.
- For decimal number 52, the number is
  - $5*10^1 + 2*10^0 = 50 + 2 = 52$
- In computers, we are concerned with four kinds of number systems, as follows:
  - Decimal Number System —Base 10
  - Binary Number System —Base 2
  - Octal Number System —Base 8
  - Hexadecimal Number System—Base 16

POSITIONAL NOTATION

- **Decimal (base 10)** numbers are expressed in the **positional notation**.

$$4202 = 2x10^0 + 0x10^1 + 2x10^2 + 4x10^3$$

Most Significant Digit

Least Significant Digit

$$4202 = 2x10^0 + 0x10^1 + 2x10^2 + 4x10^3$$

- **Binary (base 2)** numbers are also expressed in the **positional notation**.

The right-most is the least significant digit

$$10011 = 1x2^0 + 1x2^1 + 0x2^2 + 0x2^3 + 1x2^4$$

The left-most is the most significant digit

POSITIONAL NOTATION

$$10011 = 1x2^0 + 1x2^1 + 0x2^2 + 0x2^3 + 1x2^4$$

NUMBER SYSTEM

- The numbers given as input to computer and the numbers given as output from the computer, are generally in decimal number system, and are most easily understood by humans.
- However, computer understands the binary number system, i.e., numbers in terms of 0s and 1s.
- The binary data is also represented, internally, as octal numbers and hexadecimal numbers due to their ease of use.
- A number in a particular base is written as $(number)_{base}$ of number. For example, $(23)_{10}$ means that the number 23 is a decimal number, and $(345)_8$ shows that 345 is an octal number.

Quick Example

$$25_{10} = 11001_2 = 31_8 = 19_{16}$$

Base

1- Decimal Number System

- It consists of 10 digits—0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.
- All numbers in this number system are represented as combination of digits 0—9. For example, 34, 5965 and 867321.
- The position value and quantity of a digit at different positions in a number are as follows:

Position:	3	2	1	0	.	–1	–2	–3
Position Value:	10^3	10^2	10^1	10^0		10^{-1}	10^{-2}	10^{-3}
Quantity:	1000	100	10	1		1/10	1/100	1/1000

2- Binary Number System

- The binary number system consists of two digits—0 and 1.
- All binary numbers are formed using combination of 0 and 1. For example, 1001, 11000011 and 10110101.
- The position value and quantity of a digit at different positions in a number are as follows:

Position:	3	2	1	0	.	–1	–2	–3
Position Value:	2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}
Quantity:	8	4	2	1		1/2	1/4	1/8

3- Octal Number System

- The octal number system consists of eight digits—0 to 7.
- All octal numbers are represented using these eight digits. For example, 273, 103, 2375, etc.
- The position value and quantity of a digit at different positions in a number are as follows:

Position:	3	2	1	0	.	–1	–2	–3
Position Value:	8^3	8^2	8^1	8^0		8^{-1}	8^{-2}	8^{-3}
Quantity:	512	64	8	1		1/8	1/64	1/512

4- Hexadecimal Number System

- The hexadecimal number system consists of sixteen digits—0 to 9, A, B, C, D, E, F, where (A is for 10, B is for 11, C-12, D-13, E-14, F-15).
- All hexadecimal numbers are represented using these 16 digits. For example, 3FA, 87B, 113, etc.
- The position value and quantity of a digit at different positions in a number are as follows:

Position:	3	2	1	0	.	–1	–2	–3
Position Value:	16^3	16^2	16^1	16^0		16^{-1}	16^{-2}	16^{-3}
Quantity:	4096	256	16	1		1/16	1/256	1/4096

SUMMARY of NUMBER SYSTEMS

System	Base	Symbols	Used by Humans?	Used in Computers?	Largest digit
Decimal	10	0, 1, … 9	Yes	No	9
Binary	2	0, 1	No	Yes	1
Octal	8	0, 1, … 7	No	No	7
Hexa-decimal	16	0, 1, … 9, A, B, … F	No	No	F(15)

Quantities/Counting (1 of 3)

Decimal	Binary	Octal	Hexa-decimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7

Quantities/Counting (2 of 3)

Decimal	Binary	Octal	Hexa-decimal
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

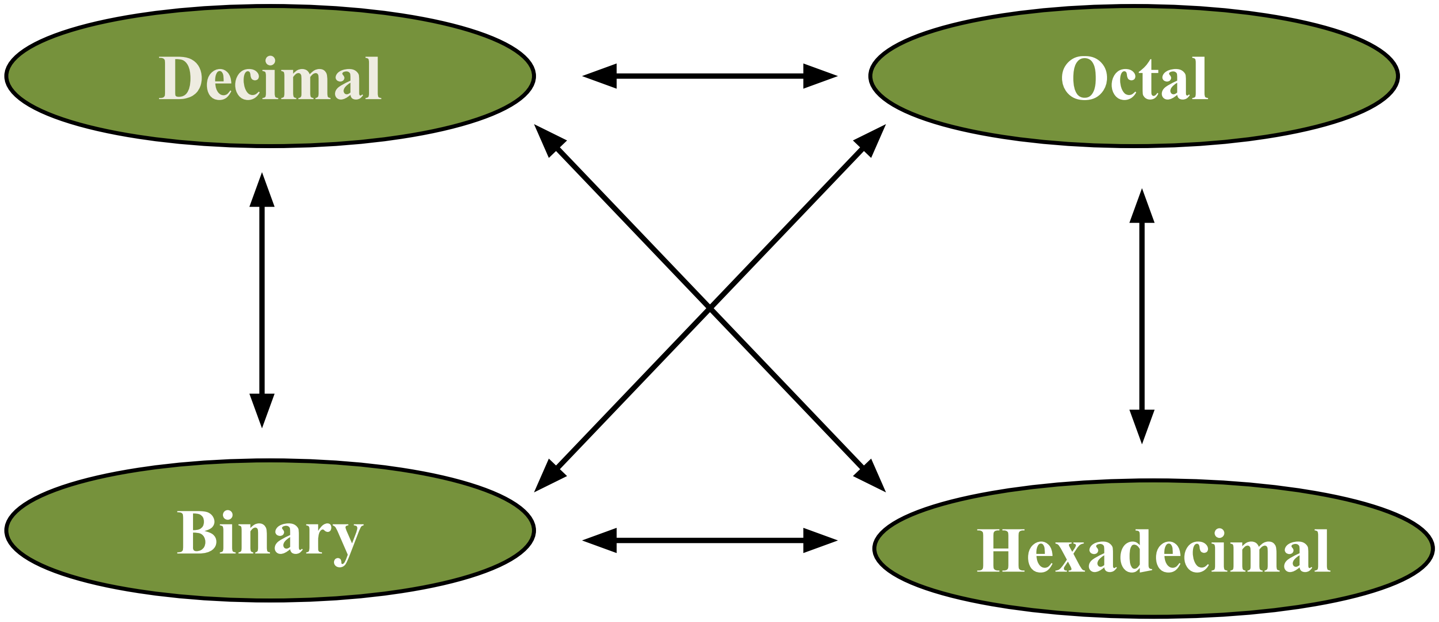
Quantities/Counting (3 of 3)

Decimal	Binary	Octal	Hexa-decimal
16	10000	20	10
17	10001	21	11
18	10010	22	12
19	10011	23	13
20	10100	24	14
21	10101	25	15
22	10110	26	16
23	10111	27	17

Etc.

CONVERSION AMONG BASES

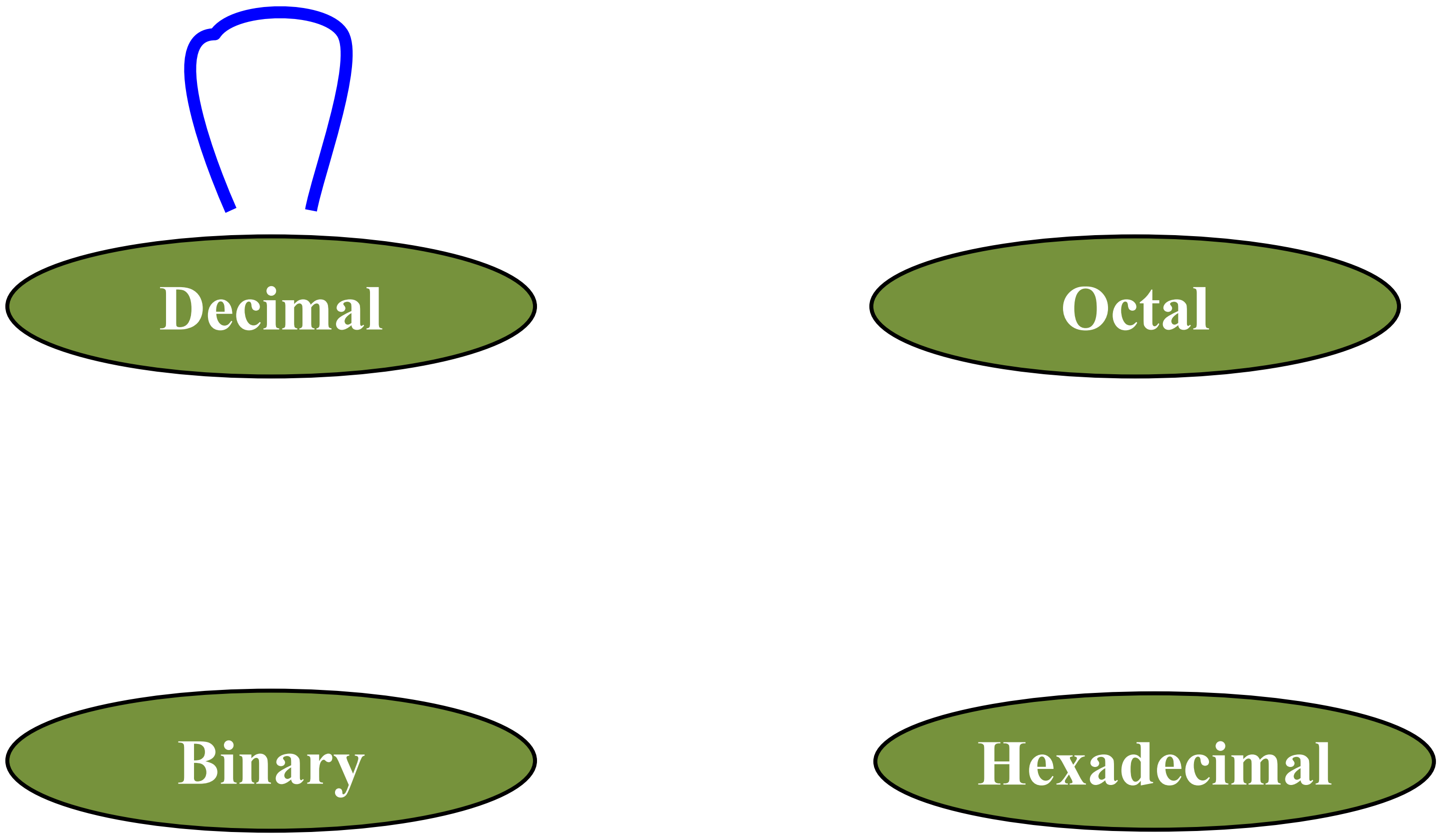
- The **possibilities:**



1- CONVERSION FROM DECIMAL other number system

- A decimal integer is converted to any other base, by using the division operation.
- To convert a decimal integer to—
  - **binary-divide by 2,**
  - **octal-divide by 8, and,**
  - **hexadecimal-divide by 16.**

1- Decimal to Decimal (just for fun)



Next slide…

Weight

$$125_{10} \Rightarrow 5 \times 10^{0} = 5$$

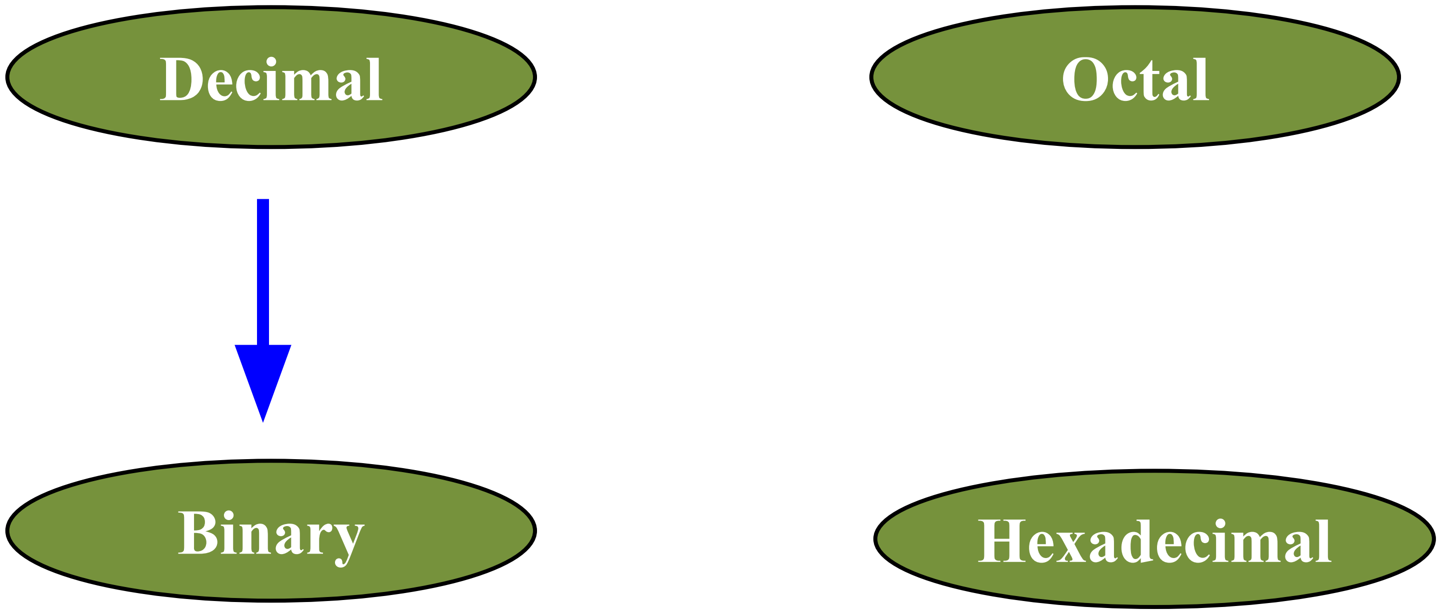
$$2 \times 10^{1} = 20$$

$$1 \times 10^{2} = 100$$

$$125$$

Base

1- Decimal to Binary



1- Decimal to Binary

- **Algorithm**
  - Divide by **two**.
  - Keep track of the **remainder**.
  - Etc.

Example

Example 1: Convert 25 from Base 10 to Base 2.

to Base	Number (Quotient)	Remainder
2	25	
2	12	1
2	6	0
2	3	0
2	1	1
	0	1

The binary equivalent of number $(25)_{10}$ is $(11001)_2$

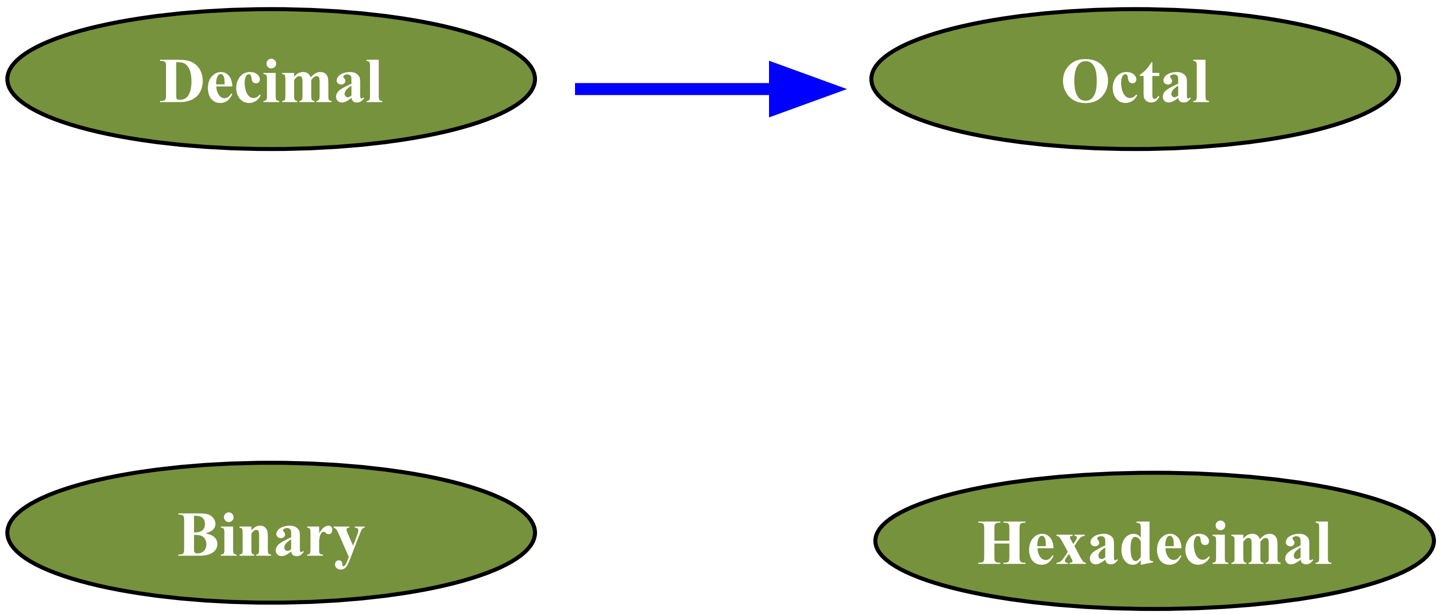
Example

$125_{10} = ?_2$

2	125	
2	62	1
2	31	0
2	15	1
2	7	1
2	3	1
2	1	1
	0	1

$125_{10} = 1111101_2$

2- Decimal to Octal



2- Decimal to Octal

- **Algorithm**
  - **Divide** by 8.
  - Keep track of the **remainder**.

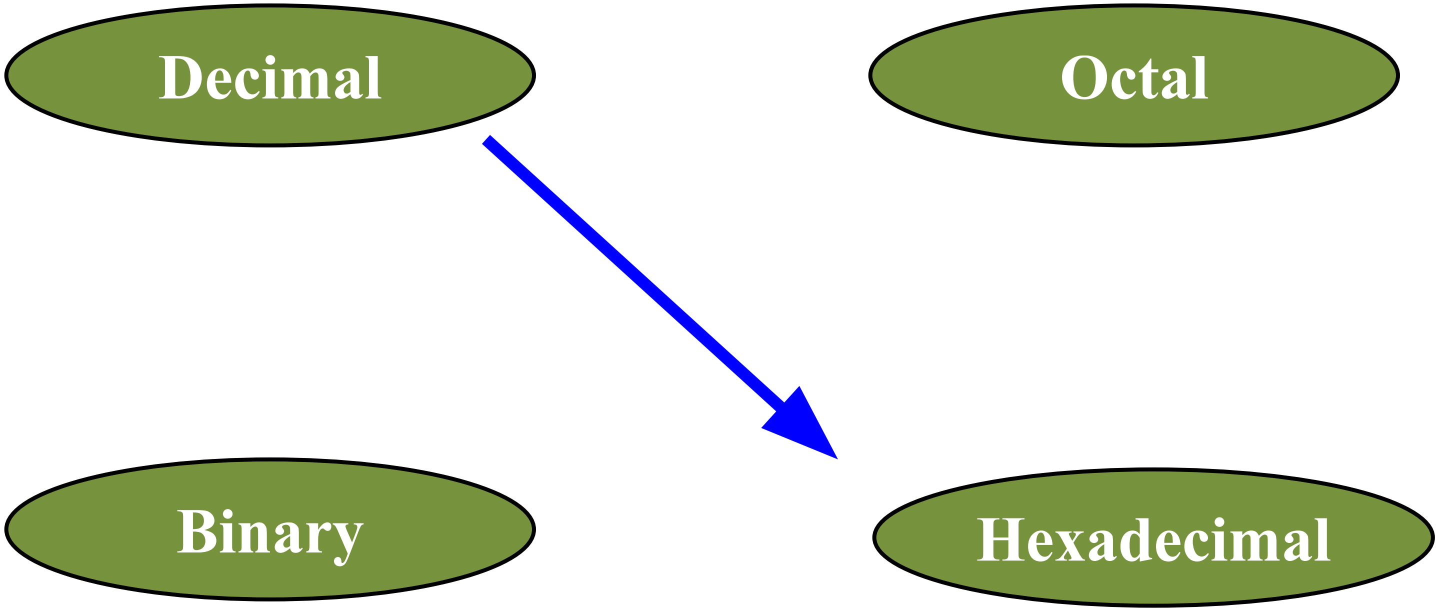
Example

$1234_{10} = ?_8$

$$
\begin{array}{r|r l}
8 & 1234 & \\ \hline
8 & 154 & 2 \\ \hline
8 & 19 & 2 \\ \hline
8 & 2 & 3 \\ \hline
 & 0 & 2
\end{array}
$$

$1234_{10} = 2322_8$

3- Decimal to Hexadecimal



3- Decimal to Hexadecimal

- **Algorithm**
  - **Divide** by 16.
  - Keep track of the **remainder**.

Example

$1234_{10} = ?_{16}$

16	1234			
16	77	2		
16	4	13	=	D
	0	4		

$1234_{10} = 4D2_{16}$

Class Tasks:

Q:1- Convert 23 from Base 10 to Base 2, 8, 16.

Q:2- Convert 147 from Base 10 to Base 2, 8 and 16.

Q:3- Convert 94 from Base 10 to Base 2, 8 and 16

Q-1

to Base	Number (Quotient)	Remainder
2	23	
2	11	1
2	5	1
2	2	1
2	1	0
	0	1

The binary equivalent of $(23)_{10}$ is $(10111)_2$

to Base	Number (Quotient)	Remainder
8	23	
8	2	7
	0	2

The octal equivalent of $(23)_{10}$ is $(27)_8$

to Base	Number (Quotient)	Remainder
16	23	
16	1	7
	0	1

The hexadecimal equivalent of $(23)_{10}$ is $(17)_{16}$

Q:2

to Base	Number (Quotient)	Remainder
2	147	
2	73	1
2	36	1
2	18	0
2	9	0
2	4	1
2	2	0
2	1	0
	0	1

The binary equivalent of $(147)_{10}$ is $(10010011)_2$

to Base	Number (Quotient)	Remainder
8	147	
8	18	3
8	2	2
	0	2

The octal equivalent of $(147)_{10}$ is $(223)_8$

to Base	Number (Quotient)	Remainder
16	147	
16	9	3
	0	9

The hexadecimal equivalent of $(147)_{10}$ is $(93)_{16}$

Q:3

to Base	Number (Quotient)	Remainder
2	94	
2	47	0
2	23	1
2	11	1
2	5	1
2	2	1
2	1	0
	0	1

The binary equivalent of $(94)_{10}$ is $(1011110)_2$

to Base	Number (Quotient)	Remainder
8	94	
8	11	6
8	1	3
	0	1

The octal equivalent of $(94)_{10}$ is $(136)_8$

to Base	Number (Quotient)	Remainder
16	94	
16	5	14
	0	5

The number 14 in hexadecimal is E.

The hexadecimal equivalent of $(94)_{10}$ is $(5E)_{16}$

Converting *Decimal Fraction* to Binary, Octal, Hexadecimal

- A fractional number is a number less than 1. It may be .5, .00453, .564, etc.
- We use the multiplication operation to convert decimal fraction to any other base.
- To convert a decimal fraction to:
  - binary-multiply by 2,
  - octal-multiply by 8, and,
  - hexadecimal-multiply by 16

Steps for conversion of a decimal fraction to any other base

1. Multiply the fractional number with the to Base, to get a resulting number.
2. The resulting number has two parts, non-fractional part and fractional part.
3. Record the non-fractional part of the resulting number.
4. Repeat the above steps at least four times.
5. Write the digits in the non-fractional part starting from upwards to downwards

Example: Convert 0.2345 from Base 10 to Base 2

$$\begin{array}{r} 0.2345 \\ \times 2 \\ \hline 0.4690 \end{array}$$

$$\begin{array}{r} .4690 \\ \times 2 \\ \hline 0.9380 \end{array}$$

$$\begin{array}{r} .9380 \\ \times 2 \\ \hline 1.8760 \end{array}$$

$$\begin{array}{r} .8760 \\ \times 2 \\ \hline 1.7520 \end{array}$$

$$\begin{array}{r} .7520 \\ \times 2 \\ \hline 1.5040 \end{array}$$

$$\begin{array}{r} .5040 \\ \times 2 \\ \hline 1.0080 \end{array}$$

The binary equivalent of $(0.2345)_{10}$ is $(0.001111)_2$

Class Task

Q:1- Convert 0.865 from Base 10 to Base 2,8 and 16.

Q-1

$$\begin{array}{r} 0.865 \\ \underline{\times 2} \\ 1.730 \\ \underline{\times 2} \\ 1.460 \\ \underline{\times 2} \\ 0.920 \\ \underline{\times 2} \\ 1.840 \\ \underline{\times 2} \\ 1.680 \\ \underline{\times 2} \\ 1.360 \end{array}$$

The binary equivalent of $(.865)_{10}$ is $(.110111)_2$

$$\begin{array}{r} 0.865 \\ \underline{\times 8} \\ 6.920 \\ \underline{\times 8} \\ 7.360 \\ \underline{\times 8} \\ 2.880 \\ \underline{\times 8} \\ 7.040 \end{array}$$

The octal equivalent of $(0.865)_{10}$ is $(.6727)_8$

$$\begin{array}{r} 0.865 \\ \underline{\times 16} \\ 5190 \\ \underline{865\ \times} \\ 13.840 \\ \underline{\times 16} \\ 5040 \\ \underline{840\ \times} \\ 13.440 \\ \underline{\times 16} \\ 2640 \\ \underline{440\ \times} \\ 7.040 \end{array}$$

The number 13 in hexadecimal is D.

The hexadecimal equivalent of $(0.865)_{10}$ is $(.DD7)_{16}$

Class Activity

Q:2- Convert 34.4674 from Base 10 to Base 2,8 and 16.

Convert 34.4674 from Base 10 to Base 2.

to Base	Number (Quotient)	Remainder
2	34	
2	17	0
2	8	1
2	4	0
2	2	0
2	1	0
	0	1

The binary equivalent of $(34)_{10}$ is $(100010)_2$

$$
\begin{array}{r}
0.4674 \\
\underline{\times 2} \\
0.9348 \\
\underline{\times 2} \\
1.8696 \\
\underline{\times 2} \\
1.7392 \\
\underline{\times 2} \\
1.4784 \\
\underline{\times 2} \\
0.9568 \\
\underline{\times 2} \\
1.8136
\end{array}
$$

The binary equivalent of $(0.4674)_{10}$ is $(.011101)_2$

The binary equivalent of $(34.4674)_{10}$ is $(100010.011101)_2$

Convert 34.4674 from Base 10 to Base 8.

to Base	Number (Quotient)	Remainder
8	34	
8	4	2
	0	4

The octal equivalent of $(34)_{10}$ is $(42)_8$

0.4674
x 8
3.7392
x 8
5.9136
x 8
7.3088
x 8
2.4704

The octal equivalent of $(0.4674)_{10}$ is $(.3572)_8$

The octal equivalent of $(34.4674)_{10}$ is $(42.3572)_8$

Convert 34.4674 from Base 10 to Base 16

to Base	Number (Quotient)	Remainder
16	34	
16	4	2
	0	2

The hexadecimal equivalent of $(34)_{10}$ is $(22)_{16}$

$$
\begin{array}{r}
0.4674 \\
\underline{\times 16} \\
28044 \\
\underline{4674x} \\
9.4784 \\
\underline{\times 16} \\
28704 \\
\underline{4784x} \\
7.6544 \\
\underline{\times 16} \\
39264 \\
\underline{6544x} \\
10.4904 \\
\underline{\times 16} \\
29424 \\
\underline{4904x} \\
7.8464
\end{array}
$$

The hexadecimal equivalent of $(0.4674)_{10}$ is $(.97A7)_{16}$

The hexadecimal equivalent of $(34.4674)_{10}$ is $(22.97A7)_{16}$

Class Tasks

Q: Convert the following decimal numbers into binary, octal and hexadecimal.

- 150.64
- 675
- 22.33
- 24.14

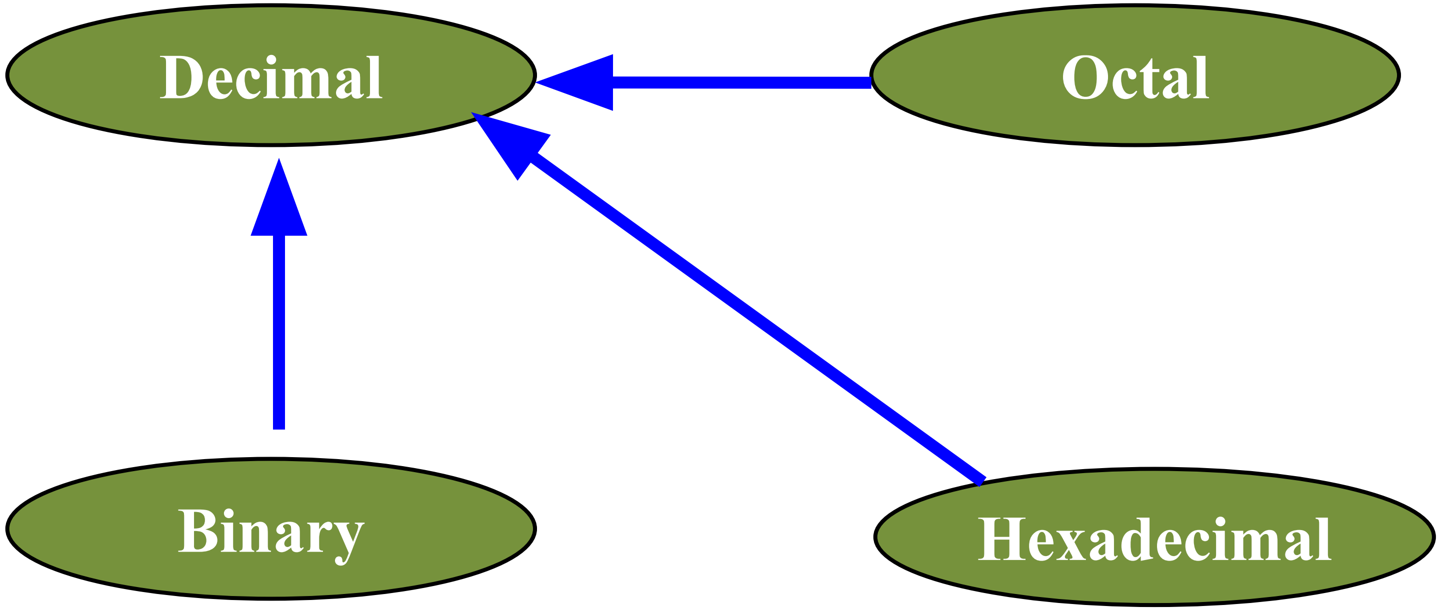
Answers

- $(150.64)_{10} = (10010110.1010)_2 = (226.5075)_8 = (96.A70A)_{16}$
- $(675)_{10} = (1010100011)_2 = (1243)_8 = (2A3)_{16}$
- $(22.33)_{10} = (10110.0101)_2 = (26.250)_8 = (16.547)_{16}$
- $(24.14)_{10} = (11000.0010)_2 = (30.1075)_8 = (18.231D)_{16}$

CONVERSION OF BINARY, OCTAL, HEXADECIMAL TO DECIMAL

- The method used for the conversion of integer part and fraction part of binary, octal or hexadecimal number to decimal number is the same; multiplication operation is used for the conversion.

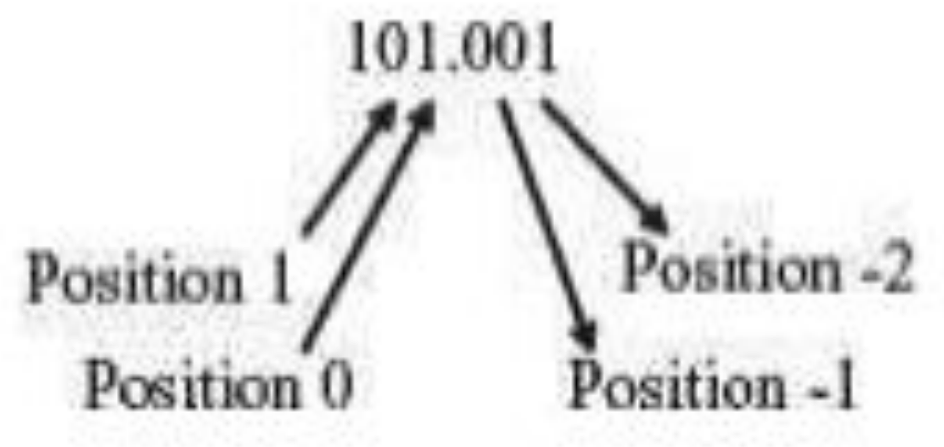
CONVERSION OF BINARY, OCTAL, HEXADECIMAL TO DECIMAL



Any number system to Decimal : multiplication with base
Decimal to Any number system : LCM/divide with base

Binary to Decimal

- **Algorithm**
  - Multiply each bit by 2^n, where n is the **"weight"** of the bit.
  - The weight is the **position** of the bit, starting from **0** on the right.
  - **Add** the results.

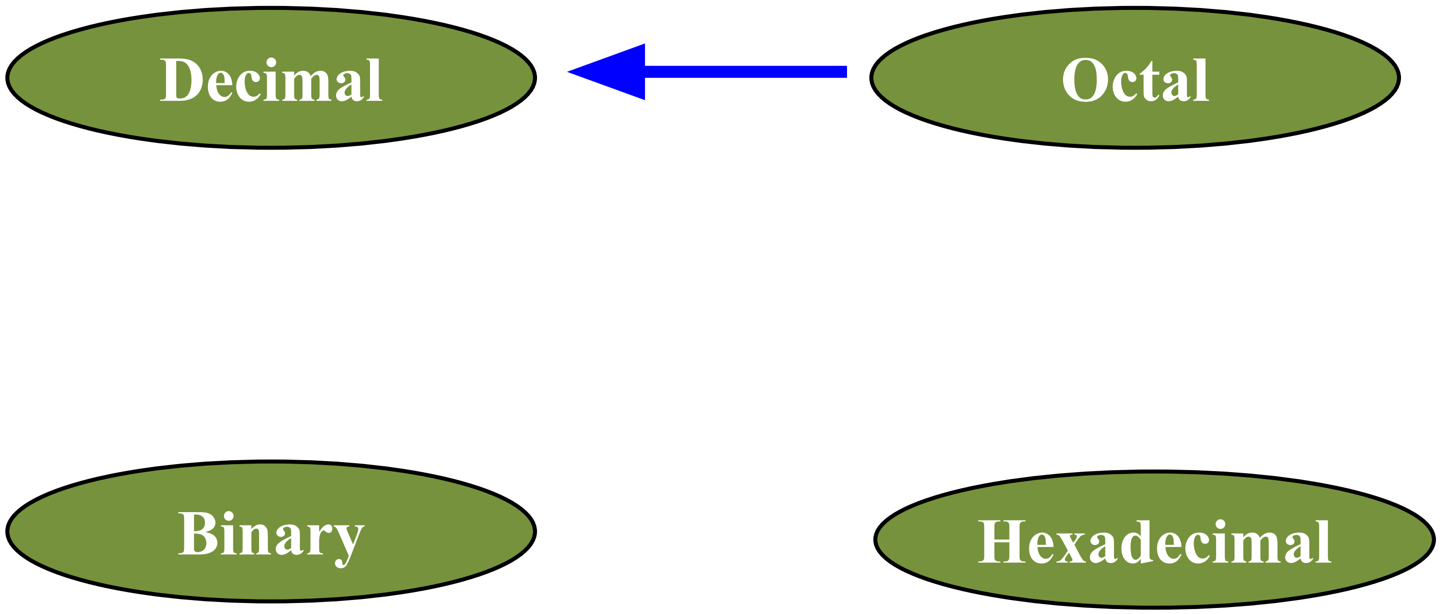


Example

Bit “0”

$$101011_2 \Rightarrow \begin{array}{rcl} 1 \times 2^0 & = & 1 \\ 1 \times 2^1 & = & 2 \\ 0 \times 2^2 & = & 0 \\ 1 \times 2^3 & = & 8 \\ 0 \times 2^4 & = & 0 \\ 1 \times 2^5 & = & 32 \\ \hline & & 43_{10} \end{array}$$

Octal to Decimal



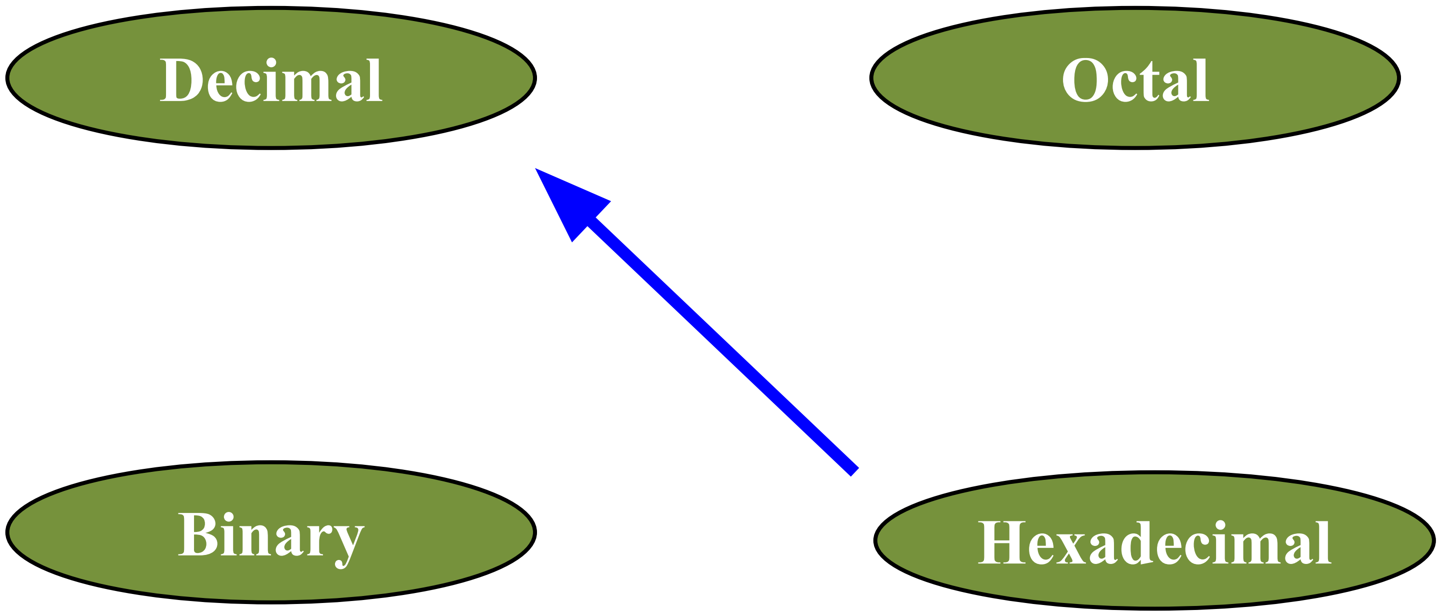
Octal to Decimal

- **Algorithm**
  - Multiply each bit by 8^n, where n is the **"weight"** of the bit.
  - The weight is the **position** of the bit, starting from **0** on the right.
  - **Add** the results.

Example

$$
\begin{array}{lrcrcr}
724_8 \Rightarrow & 4 \times 8^0 & = & & & 4 \\
& 2 \times 8^1 & = & 16 & & \\
& 7 \times 8^2 & = & 448 & & \underline{\qquad} \\
& & & 468_{10} & &
\end{array}
$$

Hexadecimal to Decimal



Hexadecimal to Decimal

- **Algorithm**
  - Multiply each bit by $\mathbf{16^{n}}$, where ***n*** is the **“weight”** of the bit.
  - The weight is the **position** of the bit, starting from **0** on the right.
  - **Add** the results.

Example

$$
\begin{aligned}
ABC_{16} \Rightarrow C \times 16^0 &= 12 \times 1 = 12 \\
B \times 16^1 &= 11 \times 16 = 176 \\
A \times 16^2 &= 10 \times 256 = 2560 \\
\hline
&\quad 2748_{10}
\end{aligned}
$$

CONVERSION OF BINARY, OCTAL, HEXADECIMAL TO DECIMAL

- Convert 1011 from Base 2 to Base 10.
- Convert 62 from Base 8 to Base 10.
- Convert C15 from Base 16 to Base 10.

1011 fromBase 2 toBase 10 $1011 = 1*2^3 + 0*2^2 + 1*2^1 + 1*2^0$ $= 1*8 + 0*4 + 1*2 + 1*1$ $= 8 + 0 + 2 + 1$ $= 11$ The decimal equivalent of $(1011)_2$ is 11.	62 fromBase 8 toBase 10 $62 = 6*8^1 + 2*8^0$ $= 6*8 + 2*1$ $= 48 + 2$ $= 50$ The decimal equivalent of $(62)_8$ is 50.	C15 fromBase 16 toBase 10 $C15 = C*16^2 + 1*16^1 + 5*16^0$ $= 12*256 + 1*16 + 5*1$ $= 3072 + 16 + 5$ $= 3093$ The decimal equivalent of $(C15)_{16}$ is 3093

CONVERSION OF (Fractional) BINARY, OCTAL, HEXADECIMAL TO DECIMAL

- Convert .1101 from Base 2 to Base 10.
- Convert .345 from Base 8 to Base 10.
- Convert .15 from Base 16 to Base 10.

.1101 fromBase 2 toBase 10	.345 fromBase 8 toBase 10	.15 fromBase 16 toBase 10
$.1101 = 1*2^{-1} + 1*2^{-2} + 0*2^{-3} + 1*2^{-4}$	$.345 = 3*8^{-1} + 4*8^{-2} + 5*8^{-3}$	$.15 = 1*16^{-1} + 5*16^{-2}$
$= 1/2 + 1/4 + 0 + 1/16$	$= 3/8 + 4/64 + 5/512$	$= 1/16 + 5/256$
$= 13/16$	$= 229/512$	$= 21/256$
$= .8125$	$= .447$	$= .082$
The decimal equivalent of $(.1101)_2$ is .8125	The decimal equivalent of $(.345)_8$ is .447	The decimal equivalent of $(.15)_{16}$ is .082

Class tasks
(Fractional and Non-fractional)

- Convert 1011.1001 from Base 2 to Base 10.
- Convert 24.36 from Base 8 to Base 10.
- Convert 4D.21 from Base 16 to Base 10.

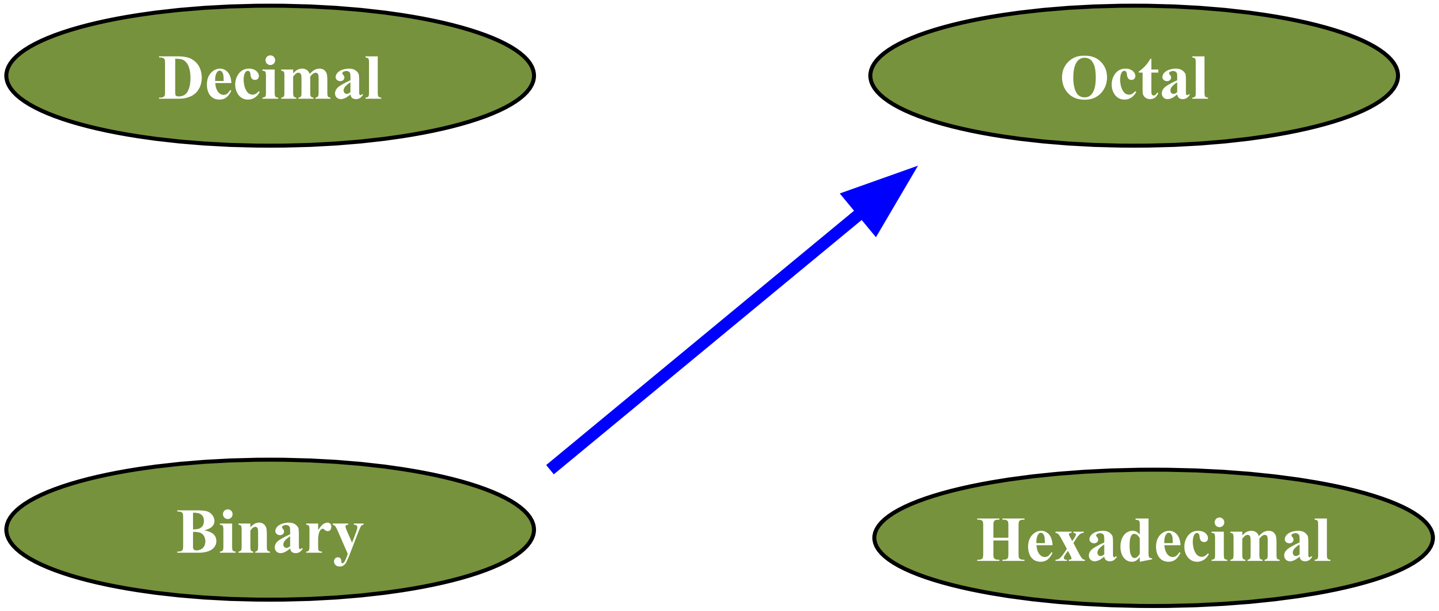
Solutions

1011.1001 fromBase 2 toBase 10	24.36 fromBase 8 toBase 10	4D.21 fromBase 16 toBase 10
$1011.1001 = 1*2^3 + 0*2^2 + 1*2^1 + 1*2^0 + 1*2^{-1} + 0*2^{-2} + 0*2^{-3} + 1*2^{-4}$	$24.36 = 2*8^1 + 4*8^0 + 3*8^{-1} + 6*8^{-2}$	$4D.21 = 4*16^1 + D*16^0 + 2*16^{-1} + 1*16^{-2}$
$= 8 + 0 + 2 + 1 + 1/2 + 0 + 0 + 1/16$	$= 16 + 4 + 3/8 + 6/64$	$= 64 + 13 + 2/16 + 1/256$
$= 11 + 9/16$	$= 20 + 30/64$	$= 77 + 33/256$
$= 11.5625$	$= 20.4687$	$= 77.1289$
The decimal equivalent of $(1011.1001)_2$ is 11.5625	The decimal equivalent of $(24.36)_8$ is 20.4687	The decimal equivalent of $(4D.21)_{16}$ is 77.1289

CONVERSION OF BINARY TO OCTAL, HEXADECIMAL

- A binary number can be converted into octal or hexadecimal number using a shortcut method. The shortcut method is based on the following information:
  - An octal digit from 0 to 7 can be represented as a combination of 3 bits, since 2 3 = 8.
  - A hexadecimal digit from 0 to 15 can be represented as a combination of 4 bits, since 2 4 = 16.

Binary to Octal



Binary to Octal

- **Algorithm**
  - Group bits in **threes**, starting on **right**.
  - **Convert** to octal digits.

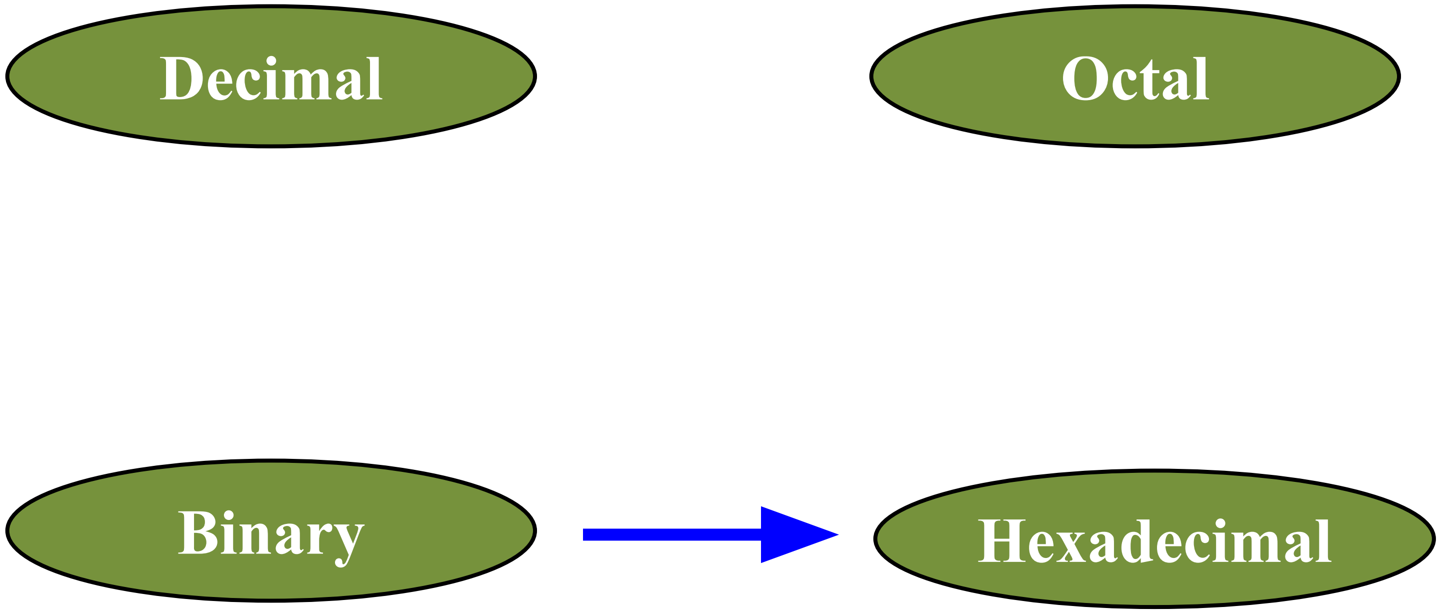
Example

$1011010111_2 = ?_8$

$$\begin{array}{cccc} 1 & 011 & 010 & 111 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 3 & 2 & 7 \end{array}$$

$1011010111_2 = 1327_8$

Binary to Hexadecimal



Binary to Hexadecimal

- **Algorithm**
  - Group bits in **fours**, starting on **right**.
  - **Convert** to hexadecimal digits.

Example

$1010111011_2 = ?_{16}$

$$\begin{array}{ccc} 10 & 1011 & 1011 \\ \downarrow & \downarrow & \downarrow \\ 2 & B & B \end{array}$$

$1010111011_2 = 2BB_{16}$

Class task

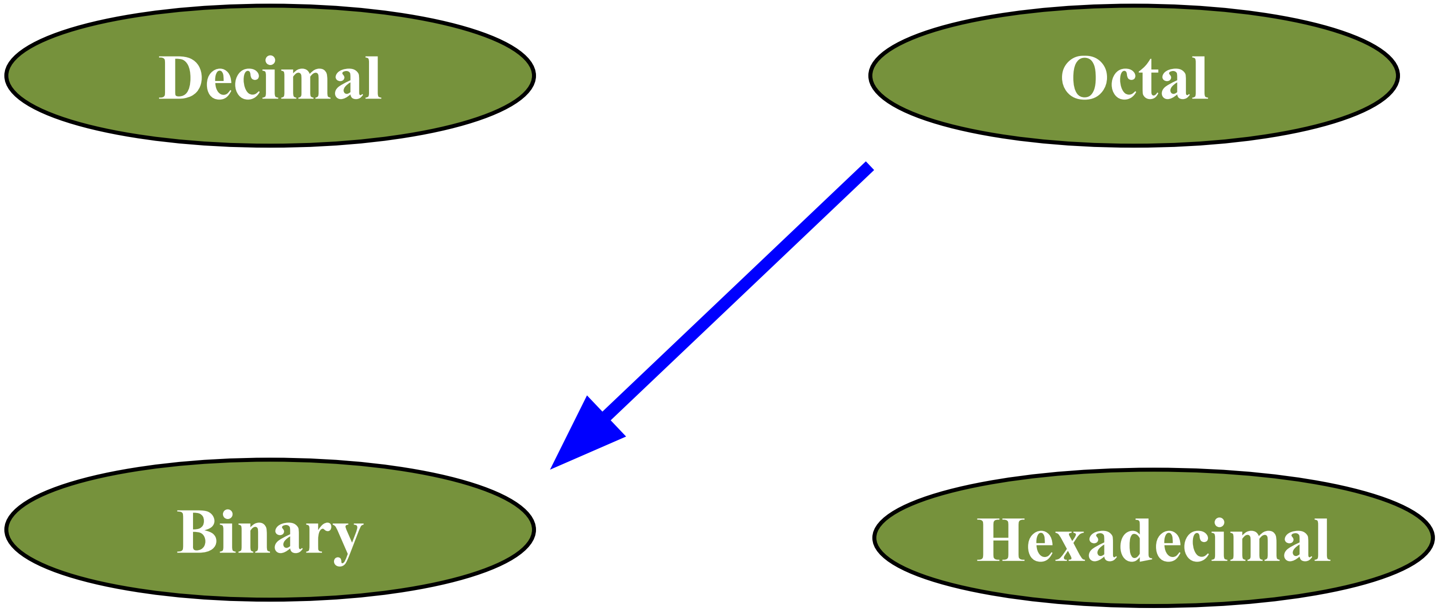
- Convert the binary number 11101011100110 to octal and hexadecimal.

- The octal number is 16546.
- The hexadecimal number is 1D66.

CONVERSION OF OCTAL, HEXADECIMAL TO BINARY

- The conversion of a number from octal and hexadecimal to binary uses the inverse of the steps defined for the conversion of binary to octal and hexadecimal.

Octal to Binary



Octal to Binary

- **Algorithm**
  - **Convert** each octal digit to a **3-bit** equivalent binary representation.

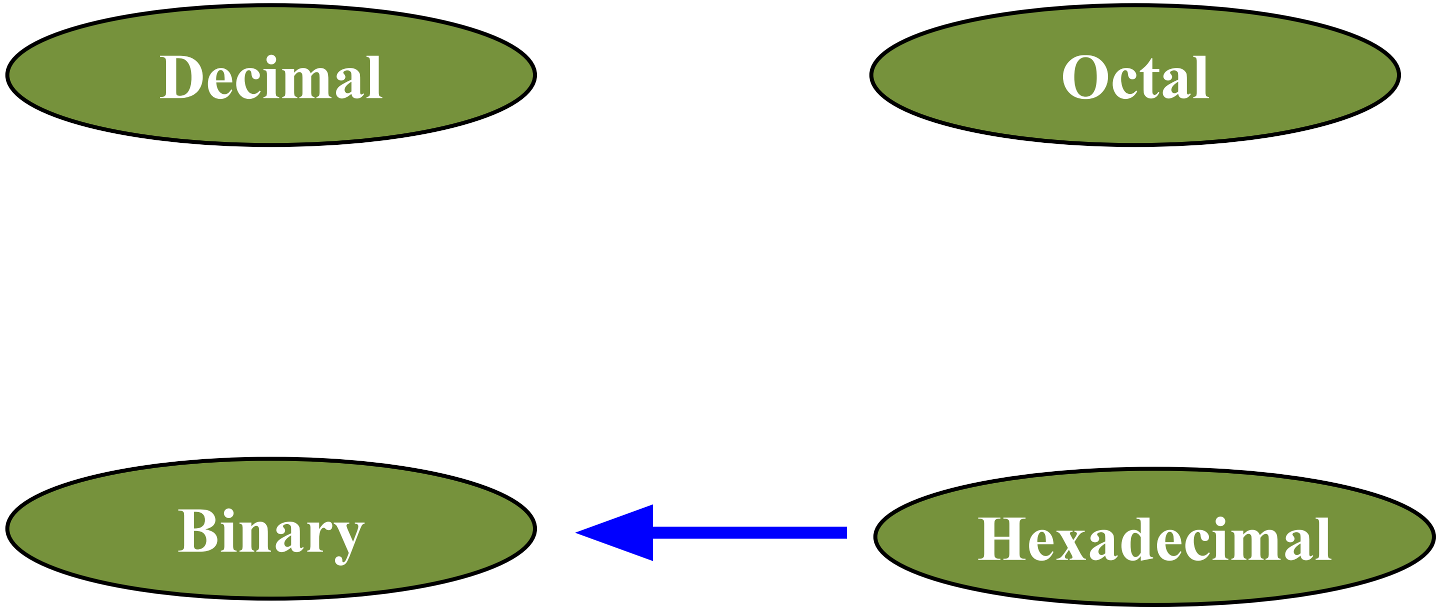
Example

$705_8 = ?_2$

$$
\begin{array}{ccc}
7 & 0 & 5 \\
\downarrow & \downarrow & \downarrow \\
111 & 000 & 101
\end{array}
$$

$705_8 = 111000101_2$

Hexadecimal to Binary



Hexadecimal to Binary

- **Algorithm**
  - **Convert** each hexadecimal digit to a **4-bit** equivalent binary representation.

Example

$10AF_{16} = ?_2$

1	0	A	F
0001	0000	1010	1111

$10AF_{16} = 0001000010101111_2$

Class Task

- Convert the octal number 473 to binary.
- Convert the hexadecimal number 2BA3 to binary.